THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — SCIENCE (086) · MID-TERM / HALF-YEARLY PRACTICE PAPER

Session 2026-27 · School Examination Pattern

Class: 9	Subject: Science	Set 05	Time: 3 Hours	Max. Marks: 80

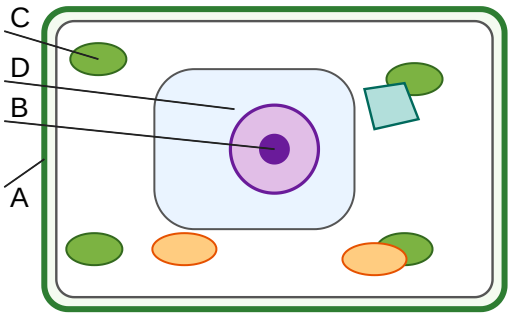
General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology (Q.1–16), Section B is Chemistry (Q.17–27) and Section C is Physics (Q.28–39).
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of the options in such questions.
3. Section A carries 32 marks, Section B 25 marks and Section C 23 marks. Marks are printed against each question.
4. Draw neat, labelled diagrams wherever required. Use g = 10 m s^{-2} wherever needed.

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. In the plant cell shown, the structure that gives the cell a definite shape and protects it is labelled: **1**



(a) A (b) B (c) C (d) D

2. Which of the following is NOT a part of the cell theory? **1**

(a) All organisms are made of cells
(b) The cell is the structural and functional unit of life
(c) All cells arise from pre-existing cells
(d) All cells contain chloroplasts

3. A gardener cuts the tip of a rose stem so that the plant becomes bushy. Further growth of that stem in length stops because the following tissue is removed: **1**

(a) apical meristem (b) lateral meristem
(c) intercalary meristem (d) collenchyma

4. Which feature distinguishes cardiac muscle from skeletal muscle? **1**

(a) Presence of striations (b) Involuntary action and branched fibres
(c) Presence of nuclei (d) Cylindrical cells

5. The insolation at a place is 0.6 kW m^{-2}. The solar energy received by 2 m^2 of area in 30 minutes is: **1**

(a) 0.6 kWh (b) 1.2 kWh (c) 0.3 kWh (d) 2.4 kWh

6. Which of the following pairs is correctly matched? **1**

(a) Monera – nucleated cells (b) Fungi – contain chlorophyll
(c) Protista – unicellular eukaryotes (d) Plantae – heterotrophs

7. In human beings, fertilisation normally takes place in the: **1**

(a) uterus (b) oviduct (fallopian tube)

(c) ovary (d) vagina

8. **Assertion (A):** Plant cells do not burst when placed in pure water. **1**

Reason (R): The cell wall of a plant cell resists the swelling of the cell.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. **Assertion (A):** Burning fossil fuels adds carbon that was locked underground to the atmosphere. **1**

Reason (R): The carbon cycle is not influenced by human activities.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. Differentiate between voluntary and involuntary muscles, with one example of each. **2**

OR

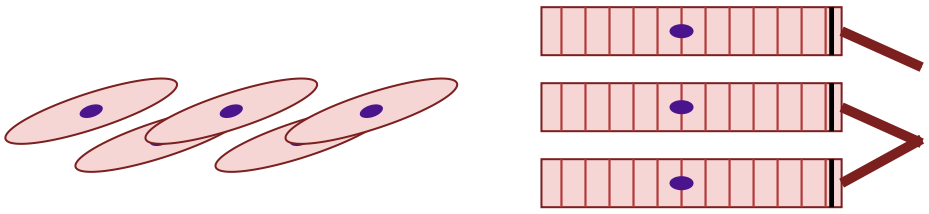
Why are cartilage and bone called connective tissues? State one function of each.

11. Describe the structure of the nucleus. State any two functions. **3**

OR

Name the cell organelle that (i) packages and dispatches materials, (ii) digests worn-out parts and (iii) synthesises lipids. Mention where each is present.

12. The figure shows two types of muscle fibres P and Q. Identify them, name one organ where each is found and state one difference between them. **3**



P **Q**

13. Fill in the blanks: **3**

Method of reproduction	Organism
Budding	(a) _____
Fragmentation	(b) _____
Spore formation	(c) _____

14. Name three greenhouse gases and write one source of each. **3**

15. A solar farm in Rajasthan has solar panels covering 50 m^2. The average insolation at the site is 0.8 kW m^{-2} and the panels receive sunlight for 6 hours a day.

(a) Calculate the solar energy received by the panels in a day. 1

(b) Why is the insolation received at the Earth's surface less than the solar constant? 1

(c) Suggest two ways of reducing the carbon footprint of a household. 2

OR

(c) How do greenhouse gases keep the Earth warm? Why is their excess harmful?

16. Differentiate between xylem and phloem on any five points. **5**

OR

Describe any four types of epithelial tissue, mentioning one location of each.

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

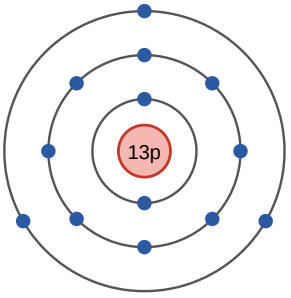
17. Which of the following mixtures will show the Tyndall effect? **1**

(a) Fog (b) Sugar solution (c) Vinegar (d) Brass

18. 5.6 g of iron combines with 3.2 g of sulphur to form 8.8 g of iron sulphide. The mass of iron sulphide formed when 11.2 g of iron combines completely with sulphur is: **1**

(a) 8.8 g (b) 11.2 g (c) 17.6 g (d) 22.4 g

19. The atom shown has 13 protons. The charge on the ion it normally forms is: **1**



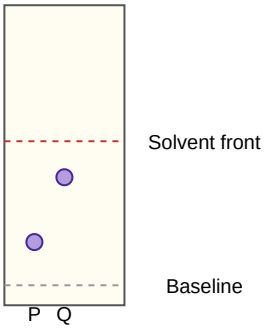
(a) 3− (b) 3+ (c) 1+ (d) 2+

20. **Assertion (A):** Paper chromatography can be used to separate the components of ink. **1**
Reason (R): Different components of ink have different solubilities in the solvent and adsorb differently on the paper.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

21. The radius of an atom is about 1×10^{-10} m and that of its nucleus is about 1×10^{-15} m. How many times larger is the atom than the nucleus? Express the answer in scientific notation. **2**

22. The chromatogram of an ink is shown. The solvent front is 80 mm from the baseline; spot P is 24 mm and spot Q is 60 mm from the baseline. Calculate the R_f values of P and Q. **2**



23. Write the chemical formula of (i) calcium hydroxide and (ii) ammonium sulphate. **2**

24. 5 g of a solute is dissolved in water to make 250 mL of solution. (i) Calculate the mass by volume percentage (% m/v). (ii) How much solute is present in 1 litre of this solution? **3**

25. State any two postulates of Dalton's atomic theory and one limitation of the theory. **3**

OR

State the law of conservation of mass. 12 g of carbon burns in 32 g of oxygen. Find the mass of carbon dioxide formed.

26. The table gives the number of sub-atomic particles in four species A, B, C and D.

Species	Protons	Neutrons	Electrons
A	6	6	6
B	6	8	6
C	7	7	7
D	11	12	10

(a) Which two species are isotopes of the same element? 1

(b) What is the mass number of species D? 1

(c) What is the charge on D? Justify. 2

OR

(c) Are B and C isobars? Give a reason.

27. Distinguish between electron, proton and neutron on the basis of (i) charge, (ii) relative mass, (iii) location, (iv) symbol and (v) discoverer. **5**

OR

Describe Rutherford's α-particle scattering experiment. State three observations and two conclusions.

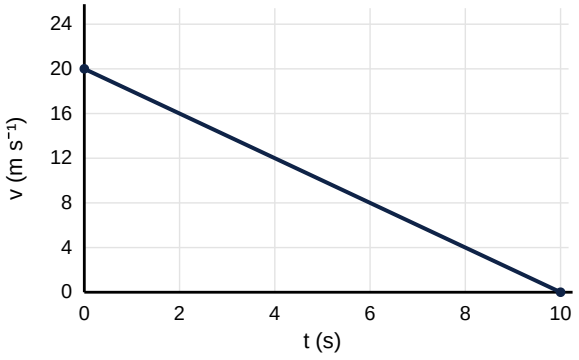
SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A car goes from A to B, 120 km apart, at 60 km h^{-1} and returns at 40 km h^{-1}. Its average speed for the round trip is: **1**

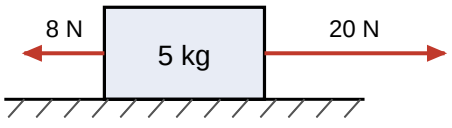
(a) 50 km h^{-1} (b) 48 km h^{-1} (c) 52 km h^{-1} (d) 100 km h^{-1}

29. The velocity–time graph of a body is shown. The magnitude of its acceleration is: **1**



(a) 0.5 m s^{-2} (b) 2 m s^{-2} (c) 10 m s^{-2} (d) 20 m s^{-2}

30. A 5 kg block is pulled by a horizontal force of 20 N against a friction force of 8 N, as shown. Its acceleration is: **1**



(a) 4 m s^{-2} (b) 2.4 m s^{-2} (c) 1.6 m s^{-2} (d) 5.6 m s^{-2}

31. A 0.2 kg ball strikes a wall at 10 m s^{-1} and rebounds at 10 m s^{-1} in the opposite direction. The magnitude of the change in its momentum is: **1**

(a) 0 (b) 2 kg m s^{-1} (c) 4 kg m s^{-1} (d) 20 kg m s^{-1}

32. A person lifts a 10 kg bag through 2 m in 4 s. The power developed is (g = 10 m s^{-2}): **1**

(a) 20 W (b) 50 W (c) 80 W (d) 200 W

33. In a longitudinal sound wave in air, the regions of high pressure are called: **1**

(a) rarefactions (b) crests (c) compressions (d) troughs

34. **Assertion (A):** A person is able to walk forward on a rough road. **1**
Reason (R): Friction between the feet and the ground provides the forward push.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

35. A 50 kg man climbs 20 stairs, each 20 cm high. Calculate the work done against gravity and the gain in his potential energy (g = 10 m s^{-2}). **2**

36. Define amplitude of a sound wave. How does it affect the loudness of sound? **2**

OR

What is reverberation? State one way to reduce it in an auditorium.

37. A body falls freely from rest (g = 10 m s^{-2}). Find (i) its velocity after 2.5 s, (ii) the distance fallen in 2.5 s and (iii) the time taken to fall through 125 m. **3**

38. A swimmer pushes the water backwards with her hands and moves forward. A student also released an inflated balloon with its mouth open and noticed that it flew away in the opposite direction to the escaping air.

(a) Which law of motion explains both observations? 1

(b) What is the reaction force when the swimmer pushes the water backwards? 1

(c) If action and reaction are equal and opposite, why do they not cancel each other? 2

OR

(c) A gun of mass 2 kg fires a bullet of mass 20 g at 100 m s^{-1}. Find the recoil velocity of the gun.

39. A source of frequency 400 Hz produces sound in air (speed 340 m s^{-1}). Find (i) the wavelength, (ii) the time period, (iii) the number of waves produced in 5 s, (iv) the wavelength in water where the speed is 1500 m s^{-1} and (v) the distance of a cliff if an echo is heard 3 s after a shout. **5**

OR

A 4 kg body is thrown vertically upwards with a velocity of 20 m s^{-1} (g = 10 m s^{-2}). Find (i) its initial kinetic energy, (ii) the maximum height, (iii) its potential energy at the maximum height, (iv) its kinetic energy at a height of 10 m and (v) its speed at that height.